

Bellcourt Review Copilot

Prior authorization under pressure | FDE hackathon | Lohith Devaramane

Bellcourt is **late because requests arrive incomplete**, and **wrong because reviewers cannot find the rule that governs**. It is not mainly a staffing problem.

I built an Agentic RAG copilot that picks the governing rules in code, checks every criterion with quoted evidence, catches incomplete requests in the first minute, and leaves every decision to a person.

97.5%

Copilot agrees with the QA auditor

117 of 120 audited cases

56.7%

Human reviewers, same cases

Contract standard is 95%

0

Wrongful approvals

and no denial is possible by design

~6 s

Per case

about a third of a cent

Live demo: bellcourt-review-copilot.vercel.app | Code: github.com/devaramanelohith/bellcourt-review-copilot

The business

Who Bellcourt is and why prior authorization matters to it

- Bellcourt is a **third-party administrator**: it runs health plans for **38 employers** who pay their staff's medical bills from their own money (453,000 people).
- Since 2023 it also decides prior authorizations for **Riverbend Health Plan** (153,700 members). Riverbend stays accountable to regulators, so it audits Bellcourt and charges penalties.
- **Prior authorization** = a check before an expensive service: is it covered under this plan, and is it medically necessary under written criteria? About 310,000 requests a year, 46% by fax.
- Nurses can approve. Only a physician can deny. Decisions have legal deadlines: 7 days for Medicare Advantage, 15 days otherwise, 72 hours if urgent.

14% to 8%

Operating margin, 3 years

Staff cost +28%, revenue +9%

\$1.9M

Penalties to Riverbend this year

About a fifth of profit

31 Dec

Riverbend contract ends

Corrective Action Plan open since July

1. The problem

What leadership sees

90.1%

Riverbend decisions on time

Standard is 97%

56.2%

Riverbend denials overturned on appeal

Review trigger is 30%

56.7%

Decisions correct in QA audit

Standard is 95%

20%

Calls abandoned

2,400 calls a day

Leadership does not agree on the cause.

Who	Their explanation
VP Operations	"It's a staffing problem, full stop." Wants 20 more nurses.
CEO	Doubts it, cannot prove otherwise. Wants "faster and more defensible, not just faster". Will not be "the next headline" on AI denials.
Riverbend	"That's a wrong-decision problem." Many denials cited the wrong criteria or an outdated version.
Senior nurse	"The clinical part is the easy part. The hard part is figuring out which rules apply."

2. How I found it: test every claim against the data

2,400 requests, 120 audited cases, 90 weeks of staffing, 3,200 calls, 38 documents, 8 interviews

Claim	What the data shows	Verdict
It is a staffing problem	Nurse count fell 57.7 to 50.6, but on-time rate does not move with it (correlation -0.05). Timeliness was already 87% when staffing was at 98% of budget.	Mostly wrong
Overturns come from new information at appeal	Requests that arrived complete are overturned more often (56.7%) than incomplete ones (48.7%).	Not supported
It is a wrong-decision problem	39 of 52 audited errors are wrong-rule errors. 34% of 2026 denials cite a policy version already retired.	Supported
A third of faxes are missing something	66 of 69 late decisions arrived incomplete. Complete requests are on time 99.8% of the time.	Supported

Every number is reproducible from the data pack files. Caveat I state openly: the pack does not say the QA sample is random, so I do not extrapolate 56.7% to all decisions.

Root cause 1: reviewers apply the wrong rule

The rule depends on the client, the service and the date of service. The tools do not track any of the three.

- **Retired policy versions.** 94 of 274 denials in 2026 cite a version that was no longer in force. Appeals against those succeed **91%** of the time.
- **A memo overrode policy.** 72 lumbar MRI requests denied under a 6-week rule after the policy changed to 4 weeks. Governance says memos cannot change criteria.
- **Plan changes never reached reviewers.** Kestrel began covering bariatric surgery in January. All 8 requests since were denied as excluded.
- **Medicare rules missed.** 20 glucose-monitor denials for Medicare members under the stricter Bellcourt rule. All 11 appeals overturned.
- **The cost in time.** 14.2 of 38 minutes per case go on finding the rule. That is 18.9 of 50.6 nurses, nearly the 20 the VP wants to hire.

Root cause 2: the clock runs out before a nurse sees the case

This is what causes the \$1.9M in penalties

- Riverbend's clock starts when the **fax arrives**. Bellcourt's system starts it when someone **types the fax in**, 17 hours later on average.
- A third of faxes are missing something. Nobody tells the provider until a person has opened the image.
- In late cases, **122 of 199 hours** were spent waiting for missing information. Nurse review took about 40 hours whether the case was late or not.
- What-if on the real data: typing faxes in within an hour and shortening the wait by a quarter lifts on-time from 90.3% to **97.7%**. Cutting nurse time by a third reaches only 94.0%.

66 of 69

Late decisions arrived incomplete

99.8%

On time when the request arrives complete

17 h

Fax waits before anyone keys it

Priorities: what to fix first

Ranked by money, contract risk, client impact and fit to the CEO's brief

#	Problem	Impact	Fix	Pattern
1	Wrong rule applied	Riverbend accuracy and audit exposure. 19 nurses' worth of time. The CEO's headline risk.	Governing-rule and criteria copilot	Agentic RAG . Built
2	Clock lost at intake	Direct cause of the \$1.9M penalties and the Corrective Action Plan.	Read the fax on arrival, check completeness, draft the fax back	AI agent step. Built as a slice
3	No visibility	67% of calls ask where a request is. Harlan (\$8.1M a year) wants reporting.	Status lookup, requirements assistant, dashboard	Plain RAG. Next

Why 1 first: it is the only fix that makes Bellcourt **both faster and more defensible**; it is the cause Riverbend named; and its rule engine is the foundation for 2 and 3. Because 2 drives the penalties, I built its highest-value slice too.

3. Approach: why Agentic RAG, with code choosing the rules

RAG, Agentic RAG or an AI agent?

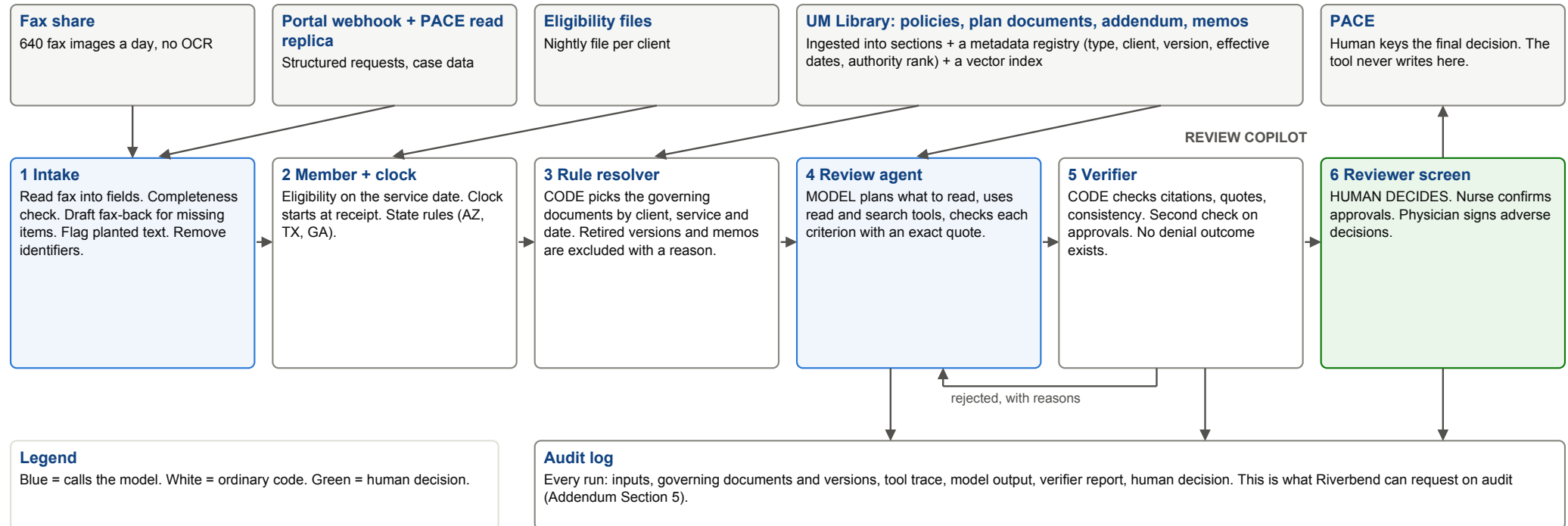
Option	What it would do here	Verdict
Plain RAG	Search by similarity and answer. But version 1 and version 2 of a policy are almost the same text, and the void memo looks highly relevant. It would repeat today's error.	Not enough
Autonomous AI agent	Decide and act on its own. The Riverbend contract and state laws forbid any tool from denying, delaying or modifying a request.	Not allowed
Agentic RAG + deterministic resolver	Code selects the documents in force from a registry (client, version, effective dates, authority rank). The model plans what to read, follows cross-references, and checks criteria. Code verifies. A person decides.	Built

- **The model never chooses the policy version.** It can only read documents the registry says are in force. Asking for a retired version is refused.
- **Evidence must be quoted.** Every MET or NOT MET needs exact words from the record. Silence means ask for information, not deny.
- **Four outcomes, none is a denial.** A skeptical second pass checks every approval.

4. Architecture

Runs beside PACE and never writes to it. Blue = model, white = code, green = human

BELLCOURT SYSTEMS (READ-ONLY). PACE STAYS THE SYSTEM OF RECORD.



Fax share, portal webhook, PACE read replica and eligibility files plug in read-only at steps 1 and 2. The human decides at step 6 and records the decision in PACE as today.

How each kind of request is read, and where the human decides

One pipeline for every channel

Channel	What arrives	How the copilot reads it
Fax (46%)	An image on the fax share. Nobody has typed anything.	A vision model transcribes the form into fields and detects blanks. The fax timestamp starts the clock.
Portal (31%)	Structured fields and notes through the portal webhook.	No reading needed. Straight to the completeness check.
Phone (15%)	A call. The agent keys the request during the call.	The agent types into the New request screen. The tool does not listen to calls. The agent sees what is missing while the caller is on the line.
Electronic (8%)	Standard X12 278 transaction; PACE creates the case.	Read from the PACE read replica as structured fields.

- **Incomplete?** A fax back to the provider is drafted in the first minute, listing exactly what to send. Intake reviews and sends it.
- **Human review.** Worklist sorted by time left. Nurse confirms approvals. Only a physician can sign an adverse decision. Roles come from sign-in and are enforced on the server.

Evidence it works

Tested on Bellcourt's own audited cases, the live queue, and hostile inputs

117/120

Agrees with the QA auditor

Humans: 68/120

29/30

Open queue incl. 13 faxes

vs my answer key

23/24

Adversarial, incomplete, wrong inputs

The 1 failure went to manual review

30/30

Unit tests, no model needed

Resolver, clock, verifier, login

Where humans went wrong	Copilot correct
A memo overrode the policy	18 of 18
Retired policy version applied	14 of 14
Client plan rule missed	6 of 7
Clinical facts misread	10 of 10

- **Planted instructions:** 10 of 10 flagged and ignored, plus the two real faxes in the queue.
- **Policy time-travel:** same record, different service date, correct version each time.
- **Zero wrongful approvals** and zero denials on the 120 audited cases.
- **Honest limits:** results vary slightly between runs; prompts were tuned on these cases, so a fresh audit sample is the next test.

Compliance and safety by design

"Show me how any tool would work under those rules" (Riverbend oversight)

Rule	How the design meets it
No tool may deny, delay or modify a request (Riverbend addendum, state laws)	No denial outcome exists. The fax back is a draft a person sends. Tests assert zero denials.
60 days' written notice before any AI tool is used	Notice sent on day 0. Pilot on self-funded employer plans while the notice runs.
Produce inputs and outputs for any case on audit	Every run logs inputs, governing documents and versions, tool trace, output, verifier report and the human decision.
Arizona: licensed medical director signs denials. Texas: disclose AI use	Flags route Arizona cases to a licensed director and add the Texas disclosure to the letter draft.
Patient data is protected	Name, date of birth, phone, record number and member ID are removed before the review step. Sign-in required; roles enforced on the server.
Provider content is untrusted	Text in faxes and notes is data. Planted instructions are flagged, cannot count as evidence, and do not change the outcome.

Deploying at Bellcourt in 90 days

PACE stays. Four people in IT. No new headcount.

When	What	Exit test
Day 0	Send Riverbend the 60-day notice. Withdraw the lumbar MRI memo. Re-review the affected denials.	No technology needed
Days 1 to 30	Host in the Azure tenant that already has the patient-data agreement. Connect read-only. Load all 38 plan documents. Replay closed cases.	95% on a fresh random audit sample, zero wrongful approvals
Days 31 to 60	Pilot with 8 senior nurses and 2 intake staff on employer plans.	Nurse minutes per case down 25%
Days 61 to 90	Go live on Riverbend with Arizona routing and Texas disclosure.	97% on time, 95% audit accuracy

- **Staffing answer:** fill the 7.4 vacant nurse posts instead of adding 20. Recovering half the rule-hunting time is worth 9.3 nurses.
- **The ask this week:** send the notice, withdraw the memo, approve the re-review.

5. Live demo

bellcourt-review-copilot.vercel.app

Step	Show	Point
1	Sign in as a nurse. Worklist sorted by time left	Case 8113 is an urgent fax that is already overdue and nobody had keyed it
2	Case 8100, lumbar MRI	Picks the current policy by date of service. The old version and the memo are listed as excluded, with reasons. Evidence is quoted
3	Case 8106, fax with a planted "pre-approved" note	Fax read into fields. Note flagged and ignored. Outcome rests on the criteria
4	Case 8120, missing second opinion	Need information. Fax back to the provider already drafted
5	Case 8105 as nurse, then as physician	The nurse cannot sign an adverse decision. The tool cannot deny
6	New request: change the date from 15 Dec 2025 to 5 Jan 2026	Same record, different policy version, different outcome
7	Evidence tab and audit trail	117 of 120 against 68 of 120. Every step logged